1. Which of the following is the correct syntax to declare a structure in C++?

A) structure Student { int rollNo; string name; }
B) struct Student { int rollNo; string name; };
C) struct Student { int rollNo, string name }
D) Struct Student { int rollNo; string name; }

---

2. A constructor is a special member function that is invoked automatically when:

A) An object is destroyed
B) An object is created
C) A function is called
D) A class is declared

---

3. What is the output of the following code?

```
class Counter {
    int count;
public:
    Counter(int c) : count(c) { cout << "Created: " << count; }
    ~Counter() { cout << "Destroyed: " << count; }
};
int main() { Counter c(5); return 0; }
```

A) Created: 5
B) Destroyed: 5
C) Created: 5Destroyed: 5
D) No output

---

4. Which access specifier allows members to be accessed by derived classes but NOT by external code?

A) public
B) private
C) protected
D) All of the above

---

5. In public inheritance, a public member of the base class becomes _______ in the derived class.

A) private
B) protected
C) public
D) inaccessible

---

6. Which symbol is used to access members of a structure or class through a pointer?

A) Dot (.)
B) Arrow (->)
C) Colon (:)
D) Double colon (::)

---

7. A pure virtual function is declared using:

A) virtual void func();
B) void func() = 0;
C) virtual void func() = 0;
D) pure void func();

---

8. When a class contains at least one pure virtual function, it becomes:

A) A concrete class
B) An abstract class
C) A final class
D) A static class

---

9. The "this" pointer in C++:

A) Points to the base class
B) Points to the current object
C) Points to the next object
D) Is a global pointer

---

10. Which of the following correctly demonstrates constructor overloading?

A) Two constructors with same parameters
B) Two constructors with different parameter lists
C) Two constructors with different return types
D) Two constructors with same name but different class names

---

11. What is the correct order of constructor execution when creating a derived class object?

A) Derived constructor → Base constructor
B) Base constructor → Derived constructor
C) Both execute simultaneously
D) Only derived constructor executes

---

12. The diamond problem in multiple inheritance is resolved using:

A) Scope resolution operator
B) Virtual base classes
C) Private inheritance
D) Function overloading

---

13. Which operator is used to access nested structure members?

A) Single dot
B) Multiple dot operators
C) Arrow operator only
D) Colon operator

---

14. What happens if you forget to define a static data member outside the class?

A) Compilation error
B) Linker error
C) Runtime error
D) No error, default value is assigned

---

15. Which inheritance type models the "IS-A" relationship most appropriately?

A) Private inheritance
B) Protected inheritance
C) Public inheritance
D) All of the above

---

16. The scope resolution operator (::) is used to:

A) Access global variables
B) Define member functions outside the class

C) Resolve naming conflicts
D) All of the above

---

17. In multiple inheritance, destructors are called in:

A) Same order as constructors
B) Reverse order of constructors
C) Alphabetical order
D) Random order

---

18. Which of the following statements about enumerations is correct?

A) enum class provides stronger type safety than plain enum
B) enum values always start from 1 by default
C) enum cannot be used in switch statements
D) enum values cannot be explicitly assigned

---

19. What is the purpose of a virtual destructor in a base class?

A) To prevent object creation
B) To ensure proper cleanup of derived objects through base pointers
C) To make the class abstract
D) To improve performance

---

20. Which of the following correctly demonstrates deep copy implementation?

A) Copying only the pointer address
B) Allocating new memory and copying actual data
C) Using the default copy constructor
D) Not copying anything

Question 1

Explain the difference between passing an object by value and passing an object by reference to a function. When would you use each approach?

Question 2

What is the Rule of Three in C++? List the three special member functions it refers to and explain when they are needed.

Question 3

Write a C++ code snippet to demonstrate constructor overloading in a `Book` class. The class should have data members: `title`, `author`, and `price`. Include at least two constructors (default and parameterized).

Question 4

What is the difference between shallow copy and deep copy? Provide a scenario where shallow copy would cause problems.

Question 5

Consider the following inheritance hierarchy: `Vehicle` (base class) → `Car` (derived class) → `ElectricCar` (further derived). Write code showing the constructor chaining when an `ElectricCar` object is created. Explain the execution order.

Question 6

What is a static data member? Give a practical example where static data members are useful (e.g., counting objects). Write the complete code for the example.

Question 7

Explain the difference between function overloading and function overriding. Provide code examples for each.

Question 8

What is the diamond problem in multiple inheritance? Draw a diagram and explain how virtual inheritance solves it.

Question 9

Write a complete C++ program to demonstrate polymorphism using virtual functions. Create a base class `Shape` with a pure virtual function `draw()`, and two derived classes `Circle` and `Rectangle` implementing this function.

Question 10

Given the following code, identify the errors and explain how to fix them:

```
class Base {
private:
    int value;
public:
    Base(int v) { value = v; }
};

class Derived : public Base {
public:
    Derived(int v) {
        value = v;  // Error 1
    }
};

int main() {
    Derived d(10);
    Base* ptr = d;  // Error 2
    return 0;
}
```

Identify both errors, explain why they occur, and provide corrected code.